

Customer Shopping Behavior Analytics Report

End-to-End Pipeline: Python ETL, PostgreSQL Data Warehousing & Power BI Dashboard Intelligence

PROJECT AUTHOR:

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Tech Stack: Python, PostgreSQL, Power BI

DATASET & PLATFORM:

Dataset Size: 3,900 Customer Transactions
Database: PostgreSQL (customer_behavior)

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1. Executive Summary

This report delivers an end-to-end analytical investigation into customer shopping behaviors, product category preferences, demographic purchase patterns, and subscription monetization across 3,900 retail transactions. Using a modern data pipeline integrating Python for exploratory data analysis (EDA) and feature engineering, PostgreSQL for structured relational data warehousing, and Power BI for interactive dashboard visualization, this audit identifies high-value customer cohorts and uncovers key growth opportunities for retail optimization.

Key headline metrics indicate that total customer spending averages \$59.76 per order with a healthy average review rating of 3.75 out of 5.0. However, only 27% of customers currently maintain an active subscription (1,053 subscribers vs 2,847 non-subscribers), highlighting a substantial untapped subscription conversion funnel. Clothing stands as the dominant revenue generator (~\$104K revenue, ~1,737 units sold), while Young Adults represent the single largest demographic sales volume driver.

Core Metric / Dimension	Observed Baseline Value
Total Customer Base Analyzed	3,900 Records
Average Purchase Amount per Transaction	\$59.76
Average Customer Review Rating	3.75 / 5.00
Subscription Adoption Rate	27.0% (1,053 Subscribed / 2,847 Non-Subscribed)
Top Revenue & Sales Category	Clothing (~\$104K Revenue, 1,737 Units Sold)
Top Purchasing Demographic Cohort	Young Adults (18–31 yrs) & Middle-aged (44–57 yrs)

2. Dataset Architecture & Data Pipeline

To establish a robust enterprise analytics foundation, raw transaction records were ingested, standardized, and enriched through a multi-tier technical architecture encompassing Python preprocessing, relational database loading, and interactive business intelligence reporting.

2.1 Data Cleaning & Feature Engineering in Python

- Schema Standardization: Converted all raw column headers into lowercase snake_case (e.g., 'Purchase Amount (USD)' to 'purchase_amount').
- Missing Value Imputation: Applied category-level median imputation on 'Review Rating' using Pandas transform functions.
- Demographic Binning (Age Groups): Quantile-based categorization (pd.qcut, q=4) segmenting

customers into four distinct cohorts: 'Young Adult' (18–31), 'Adult' (31–44), 'Middle-aged' (44–57), and 'Senior' (57–70).

4. Purchase Frequency Mapping: Formatted shopping cadence strings ('Weekly' -> 7, 'Fortnightly' -> 14, 'Quarterly' -> 90, 'Annually' -> 365 days) to compute numeric interval metrics.

2.2 PostgreSQL Database Integration (pgAdmin & SQLAlchemy)

The processed Pandas DataFrame was programmatically transferred into a local PostgreSQL database instance ('customer_behavior') using SQLAlchemy and psycopg2-binary. The target table 'customer' serves as a single source of truth (SSOT) allowing structured SQL querying and live connectivity to Microsoft Power BI.

Pipeline Stage	Technology / Tools	Key Deliverable / Outcome
Data Preprocessing & EDA	Python (Pandas, NumPy)	3,900 clean rows, age binnings, missing value handling
Relational Data Storage	PostgreSQL, SQLAlchemy, psycopg2	Structured 'customer' table in customer_behavior database
BI Dashboard Visualization	Microsoft Power BI Desktop	Interactive slicing across Category, Season, Subscription, Gender

3. Key Performance Indicators (KPIs)

The following core metrics establish the operational and financial baseline of the customer base:

3.9K	\$59.76	3.75	27%
Number of Customers	Average Purchase Amount	Average Review Rating	Subscription Adoption

\$233K+	Clothing	25.35	Montana
Estimated Total Revenue	Top Grossing Category	Mean Previous Purchases	Top Shopping Location

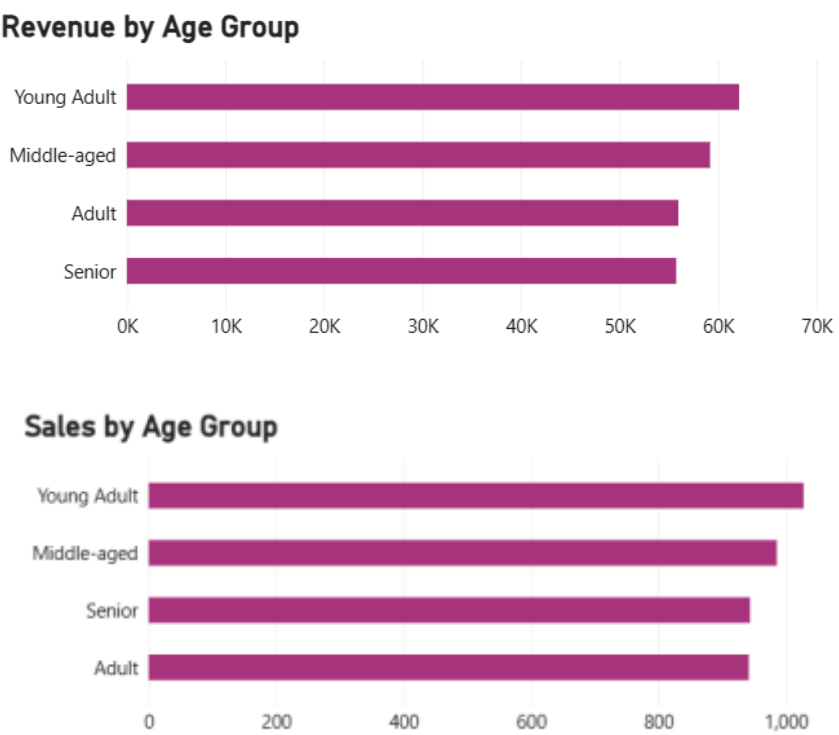
• Average Purchase Value Consistency (\$59.76)	Customer spending demonstrates strong centralization with a median of \$60.00 and standard deviation of \$23.68, showing balanced basket sizes across product lines.
• High Customer Loyalty & History (25.35 Avg Purchases)	Customers exhibit extensive historical interaction, with previous purchases ranging from 1 to 50 items (median 25), reflecting an established repeat-customer base.

• Satisfactory Brand Sentiment (3.75 / 5.0 Rating)	Review scores cluster tightly between 2.5 and 5.0 (25th percentile 3.1, 75th percentile 4.4), indicating consistent product satisfaction across all categories.
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4. Demographic & Customer Segmentation Analysis

4.1 Age Group Cohort Performance

Customers were segmented into four quartile age cohorts to analyze transaction volume and gross spend. The analysis reveals that Young Adults (18–31) generate the highest sales volume (>1,000 transactions) and top total revenue (~\$62K), closely followed by Middle-aged consumers (~\$59K revenue).



Revenue and Transaction Volume Distribution across Age Cohorts

Age Group Cohort	Age Span	Sales Volume Rank	Revenue Contribution
Young Adult	18 – 31 years	Rank 1 (Top Volume >1,000 units)	~ \$62,000 (Top Revenue Driver)
Middle-aged	44 – 57 years	Rank 2 (~980 units)	~ \$59,000 (Strong High-Ticket Spend)
Adult	31 – 44 years	Rank 4 (~940 units)	~ \$56,000 (Consistent Repeat Orders)

Senior	57 – 70 years	Rank 3 (~945 units)	~ \$56,000 (Stable Loyalty & Frequency)
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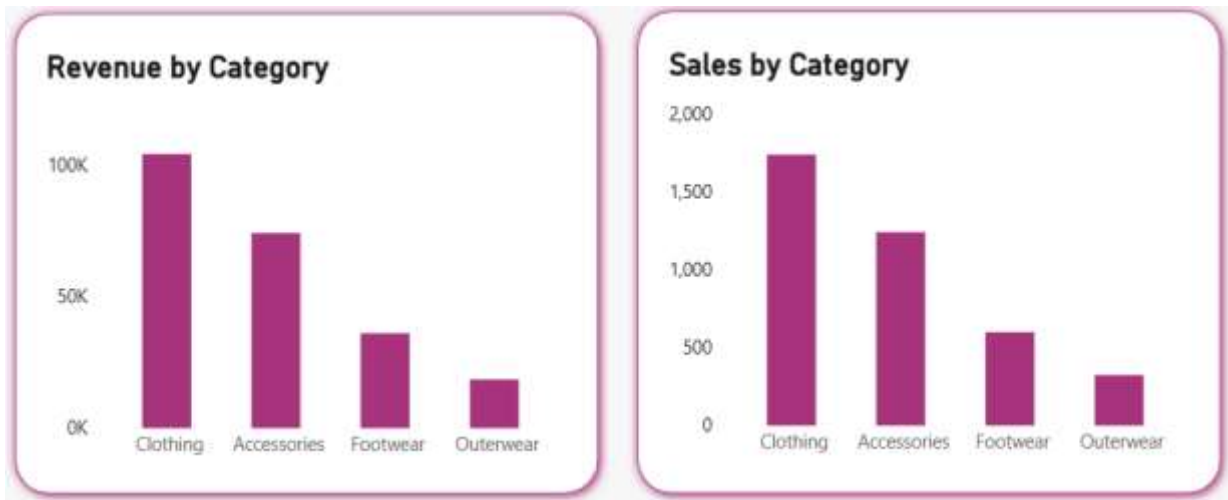
4.2 Gender Distribution & Shopping Trends

Male customers account for 68% of transactions (2,652 records) with an average ticket size of \$59.54, while Female shoppers account for 32% (1,248 records) with an average spend of \$60.25. While male shoppers drive sheer transaction volume, female shoppers demonstrate marginally higher average revenue per transaction.

Gender Revenue Disparity	Male transaction volume is more than double female volume, but basket value remains equal (\$59.54 vs \$60.25), indicating targeted female acquisition campaigns could unlock massive revenue growth.
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5. Product Category & Sales Revenue Performance

Transactions span four core departments: Clothing, Accessories, Footwear, and Outerwear. A direct comparison across revenue and sales units highlights extreme category concentration.



Comparative Analysis of Revenue and Units Sold by Product Department

Category	Total Units Sold	Estimated Total Revenue	Market Share / Strategic Impact
Clothing	1,737 units	~ \$104,000	Dominant category (>44% of total retail turnover)
Accessories	1,240 units	~ \$74,000	High-frequency add-on purchases & impulse buys

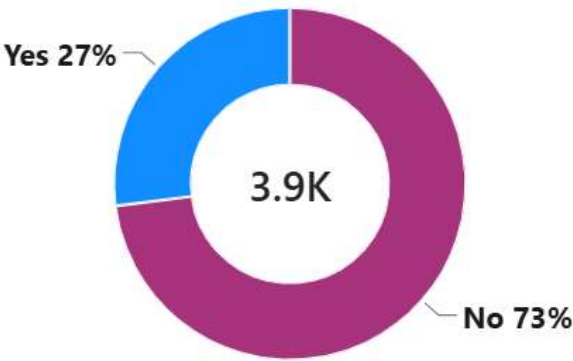
Footwear	599 units	~ \$36,000	Moderate volume with high customer review ratings
Outerwear	324 units	~ \$19,000	Seasonal concentration during Fall & Winter windows

Category Concentration Risk	Clothing and Accessories collectively account for over 76% of total revenue. Introducing bundled product strategies (e.g. Footwear + Clothing discounts) is essential to diversify category dependence.
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6. Subscription Economics & Loyalty Analysis

Subscription status serves as the primary gateway for customer lifetime value (LTV) maximization. Currently, only 27% (1,053 customers) hold an active subscription, while 73% (2,847 customers) purchase as non-subscribers. A comparative breakdown across operational metrics reveals critical behavioral parallels:

% of Customers by Subscription Status



Customer Share by Subscription status (Active Subscribers vs Non-Subscribers)

Cohort Dimension	Subscribers (Yes: 27%)	Non-Subscribers (No: 73%)	Variance / Insight
Customer Count	1,053 Users (1.1K)	2,847 Users (2.8K)	2.7x larger non-subscriber pool
Avg Purchase Amount	\$59.49	\$59.87	Equal basket spending (\$0.38 diff)

Avg Review Rating	3.75 / 5.0	3.75 / 5.0	Identical high brand satisfaction
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• Subscription Conversion Opportunity	Because non-subscribers exhibit the exact same high satisfaction (3.75 rating) and spending power (\$59.87) as subscribers, their reluctance to subscribe is likely friction in value proposition rather than dissatisfaction.
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7. Power BI Dashboard Architecture & Interactivity

To empower non-technical retail executives and regional category managers, a dedicated Power BI dashboard was developed, incorporating dynamic slicing, cross-filtering, and DAX metric formulation.



Interactive Customer Shopping Behavior Executive Dashboard in Microsoft Power BI

• Interactive Slicers Panel (Left Dock)	Provides instant multi-select cross-filtering across Subscription Status (Yes/No), Gender (Male/Female), Product Category (Clothing, Accessories, Footwear, Outerwear), and Season (Fall, Spring, Summer, Winter).
• Core Visual Matrix	1. Top Row: High-impact KPI Card blocks (Total Customers 3.9K, Avg Spend \$59.76, Avg Rating 3.75) 2. Center: Subscription Donut (27% vs 73%) alongside Category Revenue & Volume Column Charts 3. Bottom: Demographic Bar Charts mapping Revenue & Sales

	across Young Adult, Adult, Middle-aged, and Senior cohorts.
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8. Strategic Recommendations & Growth Roadmap

Based on empirical transaction analysis and dashboard findings, the following prioritised initiatives are recommended:

CRITICAL	1. Launch Target Subscription Onboarding Incentives With 73% of customers not subscribed despite identical basket spend (\$59.87), introduce a 'First-Order Subscriber Discount' or free express shipping tier. Converting 15% of non-subscribers would yield ~420 new recurring members.
HIGH	2. Implement High-Value Cross-Category Bundling Clothing generates over \$104K in revenue but Outerwear lags at \$19K. Deploy algorithmic basket recommendations at checkout (e.g., 'Pair this Sweater with Boots/Coat') to lift low-performing categories.
HIGH	3. Scale Female Customer Acquisition Channels Female shoppers generate marginally higher spend per order (\$60.25 vs \$59.54) but represent only 32% of transactions. Reallocate digital advertising spend toward female demographic channels (e.g., Instagram/Pinterest).
MEDIUM	4. Cater to Young Adult & Middle-Aged Purchasing Peaks Young Adults (>1,000 units) and Middle-aged shoppers (~\$59K revenue) represent the primary revenue engine. Align seasonal promotional calendars and trend-focused apparel inventory to these groups.
LOW	5. Automated Data Warehouse Sync & Model Integration Establish automated ETL triggers between the PostgreSQL 'customer' table and Power BI Service to enable scheduled daily data refreshes and predictive customer churn modeling.

9. Conclusion

This analytical investigation underscores a highly active, satisfied retail customer base with a robust average basket spend of \$59.76 and strong repeat loyalty (25+ previous purchases). The primary growth bottleneck lies in subscription monetization (27% adoption) and severe category concentration in Clothing.

By deploying the unified Python-PostgreSQL-Power BI analytics framework, retail decision-makers can now dynamically monitor demographic shifts, track subscription conversions, and optimize inventory allocation across seasonal cycles.

Strategic Focus Area	Current Baseline State	12-Month Target State
Subscription Adoption Rate	27.0% (1,053 Users)	> 45.0% (>1,750 Users)
Female Customer Share	32.0% (1,248 Users)	> 42.0% Share of Transactions
Outerwear & Footwear Contribution	~23.5% Combined Revenue	> 35.0% Through Product Bundling
Average Order Value (AOV)	\$59.76	> \$68.00 via Smart Upselling
Analytics Infrastructure	Manual Jupyter/Local pgAdmin	Automated Daily Power BI Service Refresh

With disciplined execution across the strategic recommendations—particularly subscription incentives and cross-category merchandising—the business can systematically elevate customer lifetime value and build sustainable revenue momentum.